



ERP Business Process, Functions and Data Requirements



OUTLINE



ERP Overview

ERP
Functional
Areas

Integrated
Information
System

Learning Objectives

01

- Define ERP and Identify its characteristics and benefits

02

- Describe the functional areas of business operations

03

- Contrast business functions and business process

04

- Explain the importance of integrated information systems in business.

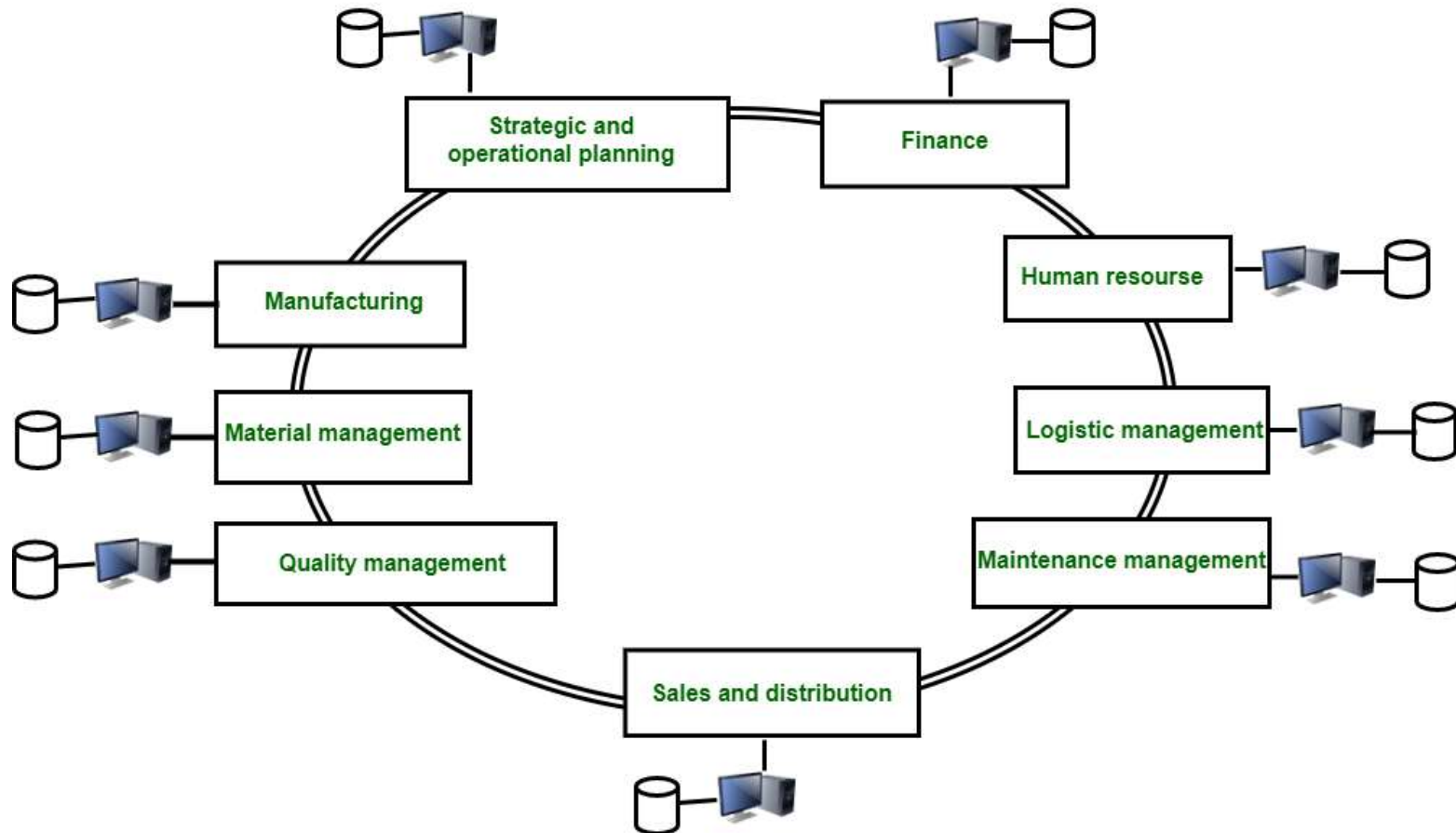
ERP Overview

Enterprise Resource Planning (ERP)

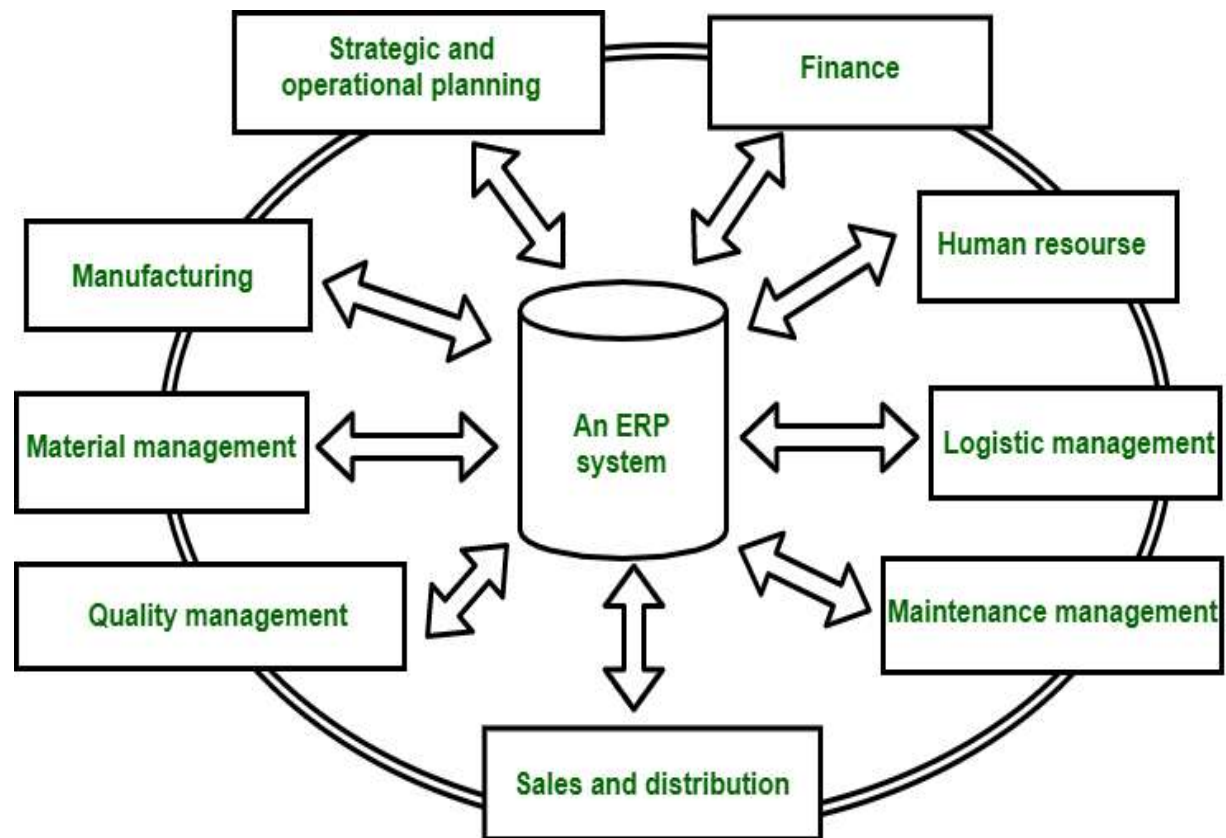
- Enterprise Resource Planning (ERP) programs are **software** **used by companies to manage information in every area of the business.**
- ERP programs help **manage company-wide business processes using a common database and shared management reporting tools.**
- ERP software supports the efficient operation of business processes by **integrating activities throughout a business.**



Before ERP



After ERP



ERP Key Characteristics

Centralized Database

- A common database for all departments
- A single source of truth for accurate and up-to-date data
- Easier access to information for everyone involved

Automated Business Processes

- Automation of repetitive and transaction-based activities
- Cost savings and zero inaccuracies
- Increased accuracy and efficiency

Consistent User Interface

- Easy-to-use interface for all employees
- Consistent look and feel for seamless collaboration
- Access to the same data from different departments

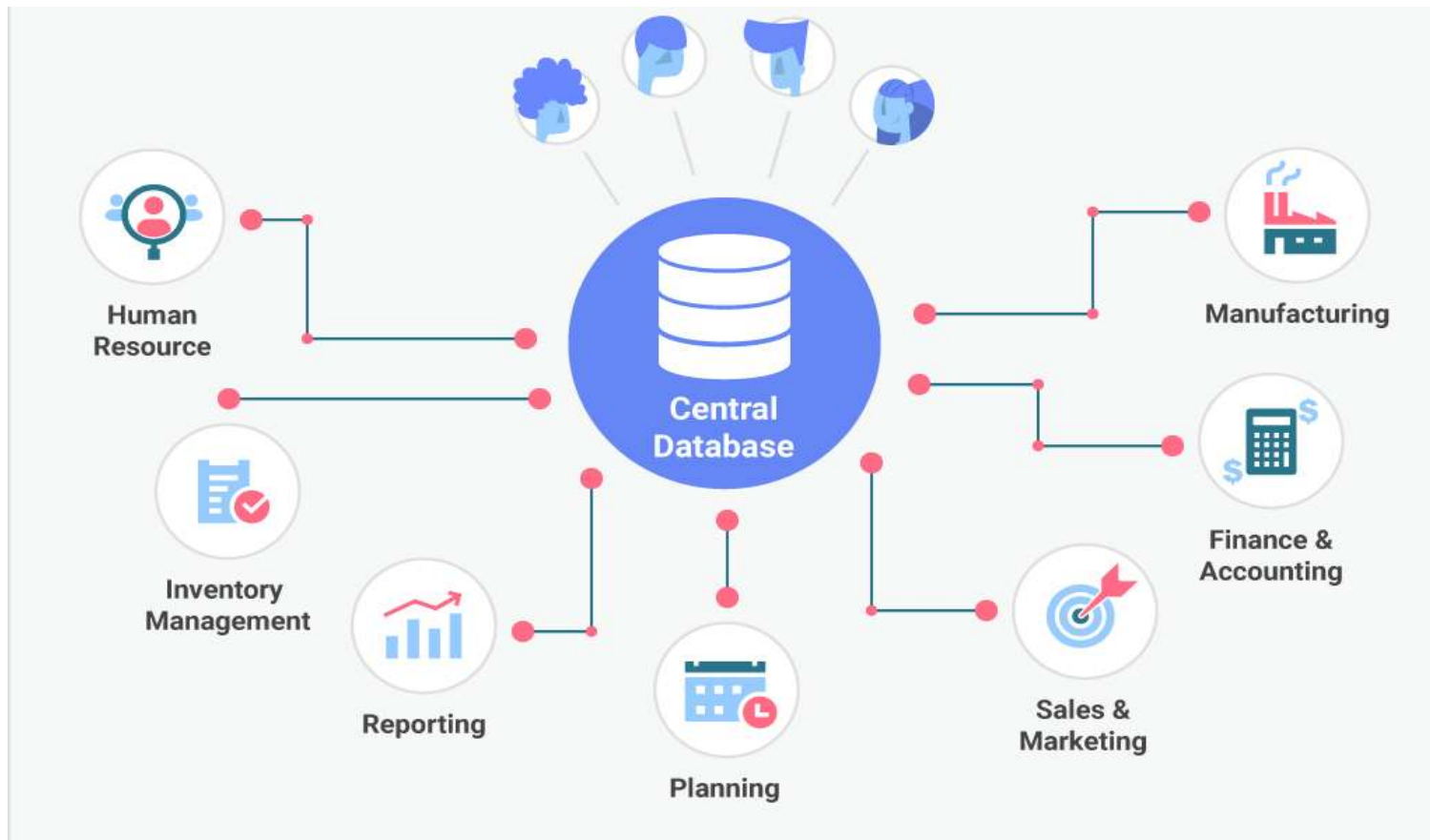
Integration of Business Functions

- Unification of different aspects of business operations
- Streamlined processes from inter-related departments
- Improved efficiency and productivity

Data Analytics

- A wealth of data for informed business decisions
- Analysis of sales data to improve revenue generation
- Examination of expense reports to reduce costs

ERP Key Characteristics (Centralised Database)



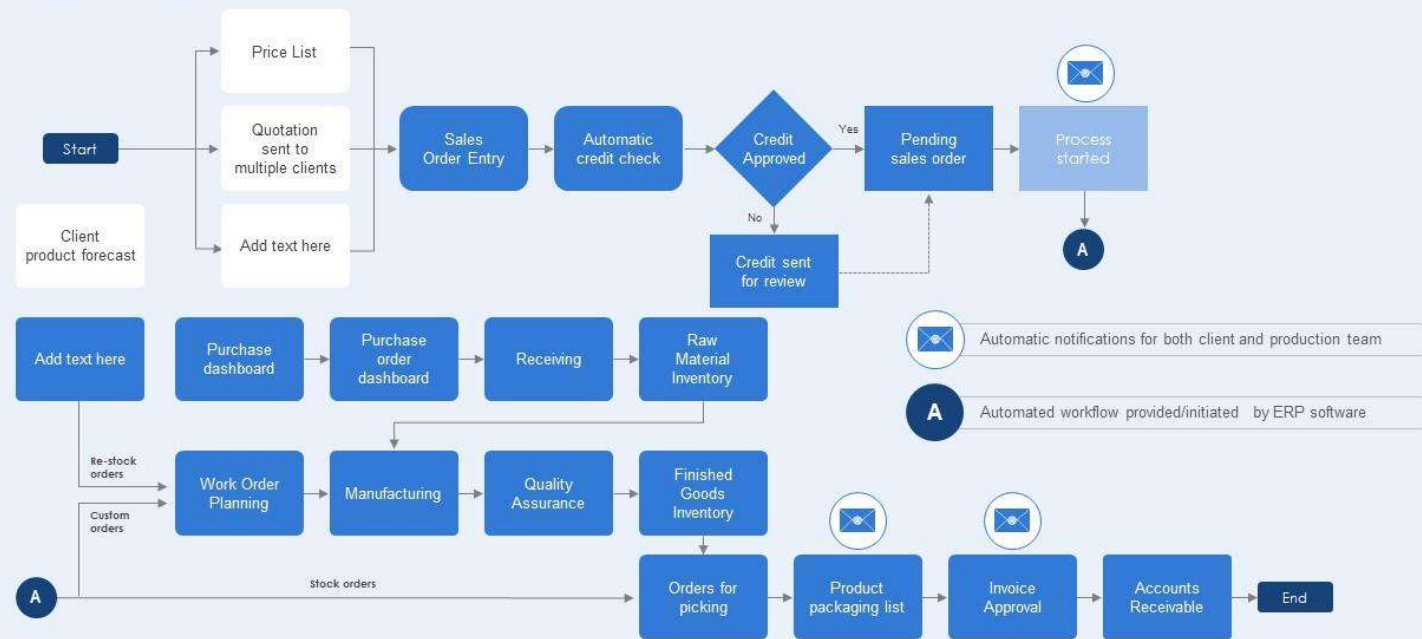
For example: PETRONAS AlphaLNG

Maintenanc department defect problem> SAP will records issue automatically>check on warehouse inventory>procurement purchase order

ERP Key Characteristics (Automated Business Processes)

Workflow and task automation through ERP

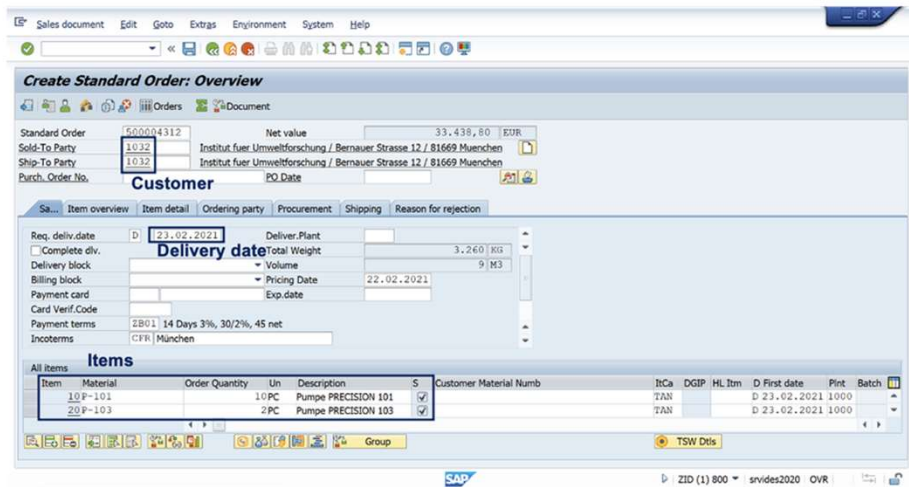
This slide showcases workflow and production tasks automation through enterprise resource planner (ERP) software. It provides information about sales order, price list, receiving, raw material inventory, invoice, purchase order, manufacturing, etc.



*ERP – Enterprise Resource Planner

This slide is 100% editable. Adapt it to your needs and capture your audience's attention.

ERP Key Characteristics (Consistent User Interface)



Create Standard Order: Overview

Standard Order: 500004312 Net value: 33,438.80 EUR

Sold-To Party: 1032 Institut fuer Umweltforschung / Bernauer Strasse 12 / 81669 Muenchen

Ship-To Party: 1032 Institut fuer Umweltforschung / Bernauer Strasse 12 / 81669 Muenchen

Purch. Order No. PO Date

Customer

Req. deliv.date: 23.02.2021 Deliver.Plant: Total Weight: 3,260 KG

Delivery block: Volume: 9 M3

Billing block: Pricing Date: 22.02.2021 Exp.date:

Payment card: Card Verif.Code: Payment terms: 2B01 14 Days 3%, 30/2%, 45 net Incoterms: CFR Muenchen

Items

Item	Material	Order Quantity	Un	Description	S	Customer Material Numb	ItCa	DGEP	HL itm	D First date	Print	Batch
10 P-101		10 PC		Pumpe PRECISION 101	✓		TAN			D 23.02.2021 1000		
20 P-103		2 PC		Pumpe PRECISION 103	✓		TAN			D 23.02.2021 1000		

SAP ZID (1) 800 - svides2020 OVR

SALES



Display Contract Elements

Find by: Person

Collective search help

Search Term

Free search

Pers. No.: 1210 Pers.Assign: 00001210 Assembly worker (D) Acti

Pers.No.: 1210 Name: Peter Bauer

EE group: 1 Active Pers.area: 1000 Hamburg

EE subgroup: DK Hourly premium wa Cost Center: 4230 Pump Assembly

Start: 01.01.1994 to 31.12.9999 Chng: 08.06.1995 LIMPET

Contractual regulations

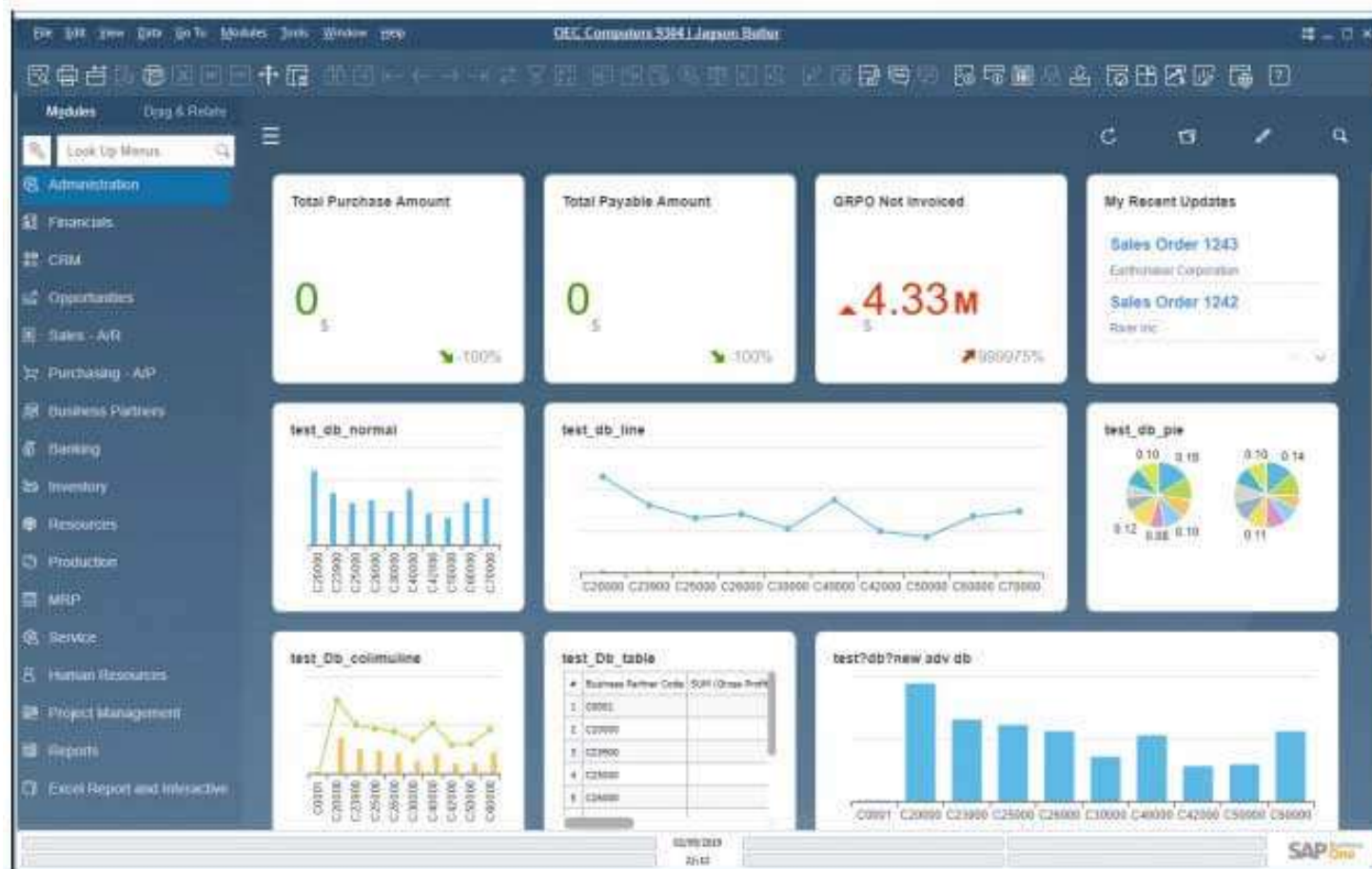
Contract Type: Unlimited

☐ Sideline job

☐ Competition clause

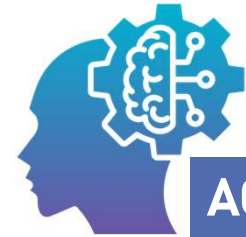
HUMAN RESOURCES

ERP Key Characteristics (Data Analytics)



ERP Benefits

Class Activity: What do you think?



ACTIVITY



What do you think
The benefits of ERP?

Functional Areas of Operation

Functional Areas of Operations

Marketing and Sales

Accounting and Finance

Supply Chain Management

Human Resources

Business Function

FUNCTIONAL AREA	Marketing and Sales	Supply Chain Management	Accounting and Finance	Human Resources
BUSINESS FUNCTIONS	Marketing of a product	Purchasing goods and raw materials	Financial accounting	Recruiting and hiring
	Taking sales orders	Receiving goods and raw materials	Cost allocation and control	Training
	Customer support	Transportation and logistics	Planning and budgeting	Payroll
	Customer relationship management	Scheduling production runs	Cash-flow management	Benefits
	Sales forecasting	Manufacturing goods		Government compliance
	Advertising	Plant maintenance		

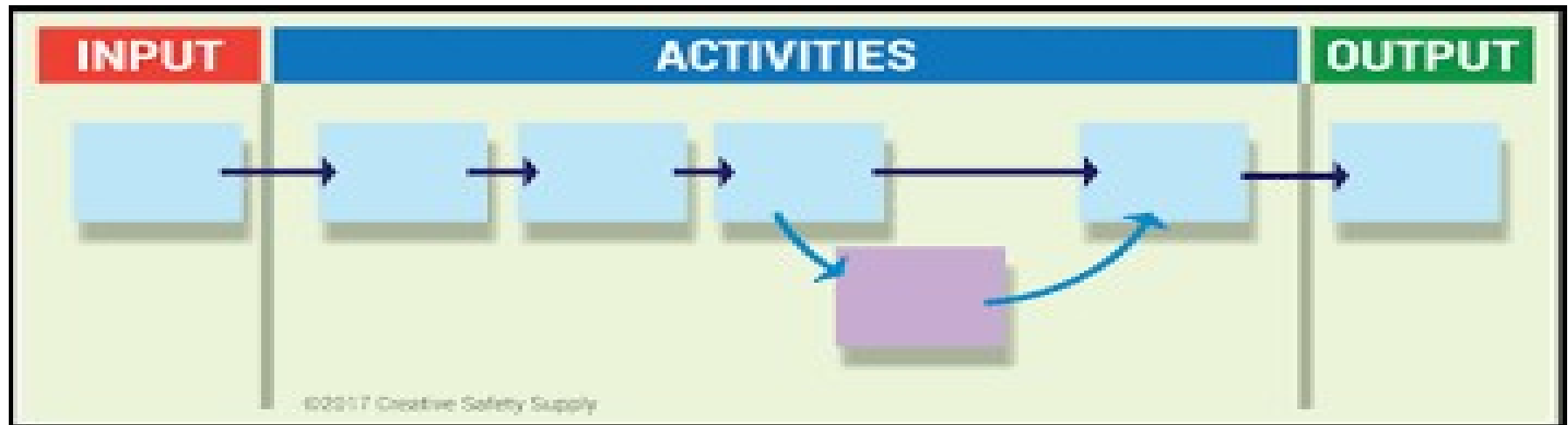
A business function is a specialized area or department within an organization that performs a specific set of related tasks or responsibilities to support the overall operation of the business.

FIGURE 1-1 Examples of functional areas of operation and their business functions

Business Process

- A **business process** is a **sequence of interrelated tasks or activities** that produce a specific service or product to achieve a business goal often **crossing multiple functions**.
- The business process view is the **customer's perspective**.
 - The customer may be the traditional external customer who buys the product or service, or an internal customer (a colleague in another department)
 - The customer does not care that different functions are involved in processing their order, and will not tolerate mistakes and delays caused by poor coordination of business functions

Business Process

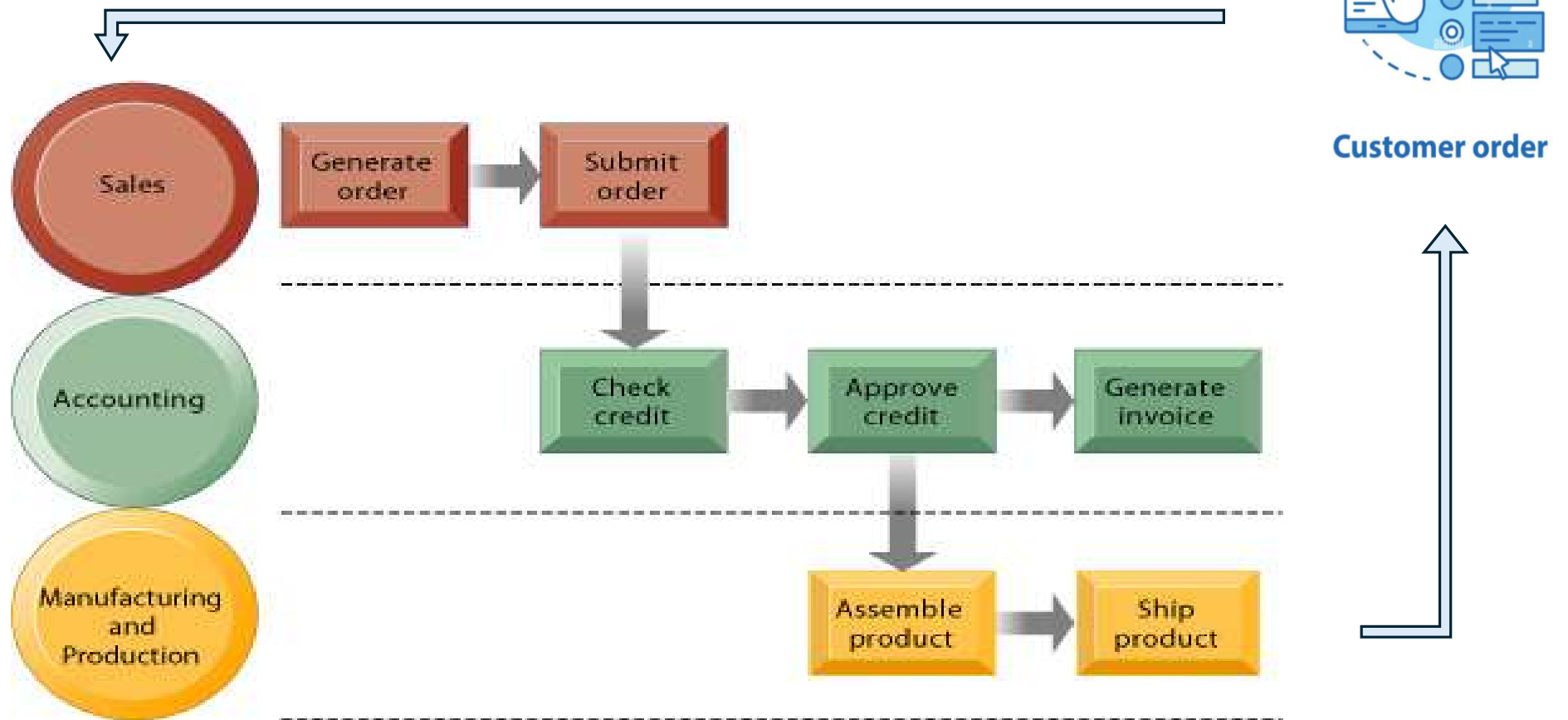


is a collection of activities that takes one or more kinds of input and creates an output that is of value to the customer.

Business Process (example)

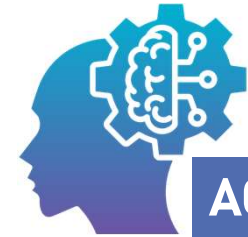
Input	Functional area responsible for input	Process	Output
Request to purchase smartphone	Marketing and Sales	Sales order	Order is generated
Financial help for purchase	Accounting and Finance	Arranging financing in-house	Customer finances through the smartphone company
Fulfillment of order	Supply Chain Management	Shipping and delivery	Customer receives smartphone
Technical support	Marketing and Sales	24-hour help line available	Customer's technical query is resolved

Business Process (example)



Business Process

Class Activity: Business Process You've Experienced



ACTIVITY

Instructions:



Please reply in the **Ms Teams** chat with:

1. One business process you've personally experienced (as a student, customer, or worker)
 2. The departments or functions involved
 2. *Example:*
 2. *Process: Registering for university course*
 3. *Functions Involved: Academic Affairs, Registrar, IT*
- You can also react or reply to your friends' answers!

Business Function vs Business Process

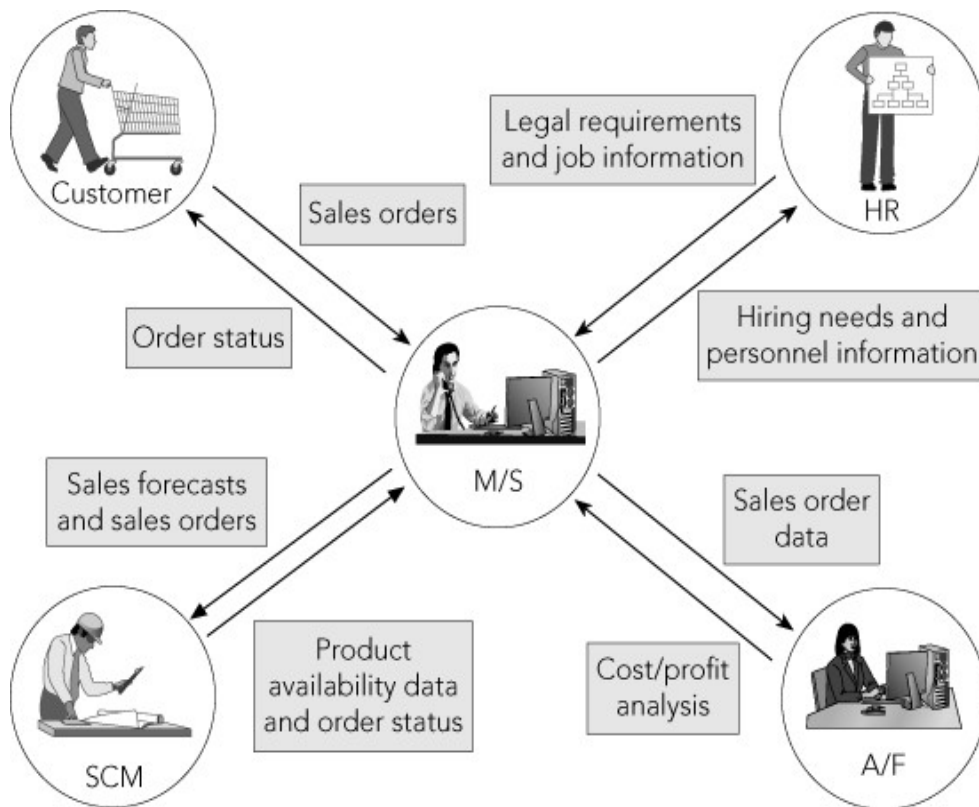
Aspect	Business Function	Business Process
Focus	Department or role	Workflow or activity flow
Boundaries	Usually stays within one department	Often spans multiple departments
Goal	Performs specialized tasks	Achieves a business outcome
Example	HR department managing employee records	Onboarding a new employee (HR + IT + Payroll)

Integrated Information System

Integrated Information System

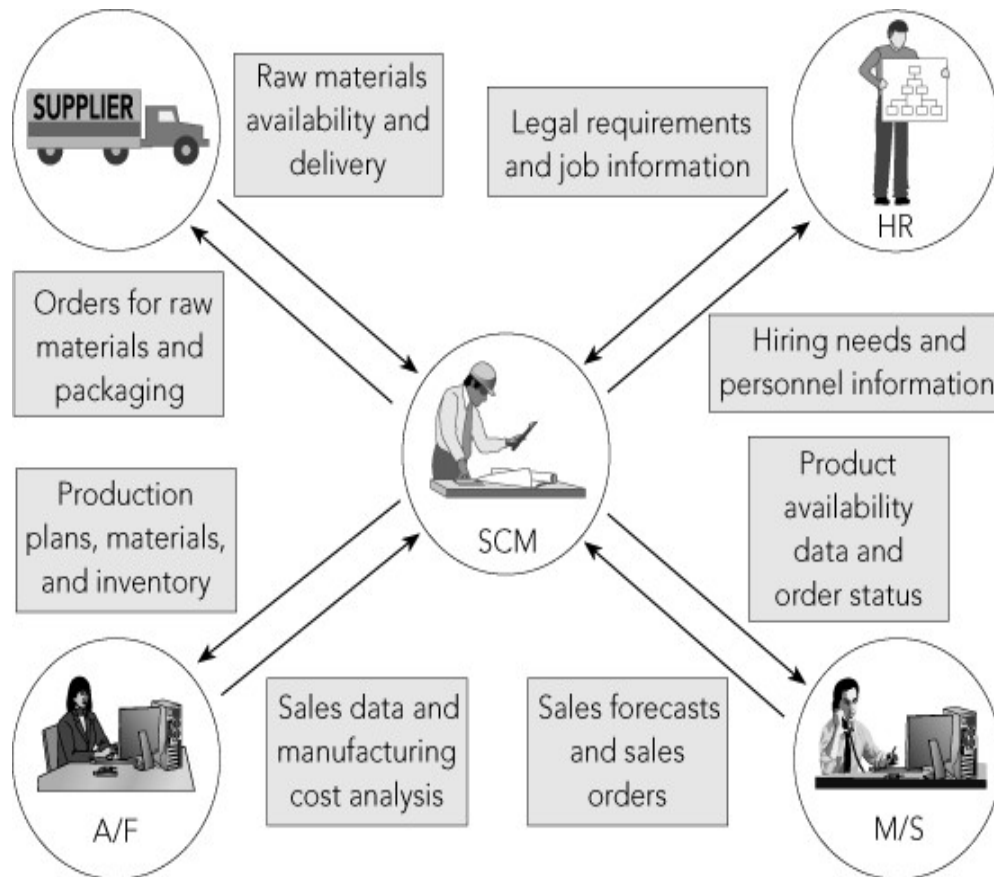
- Sharing data efficiently and effectively within and between functional areas leads to more efficient business processes
 - Functional areas are interdependent
 - Each requires data from the others
- Information systems that share data between functional areas are called **Integrated Information Systems**
- They enable real-time data sharing, streamline workflows, and support better decision-making across departments.

Marketing and Sales (M/S)



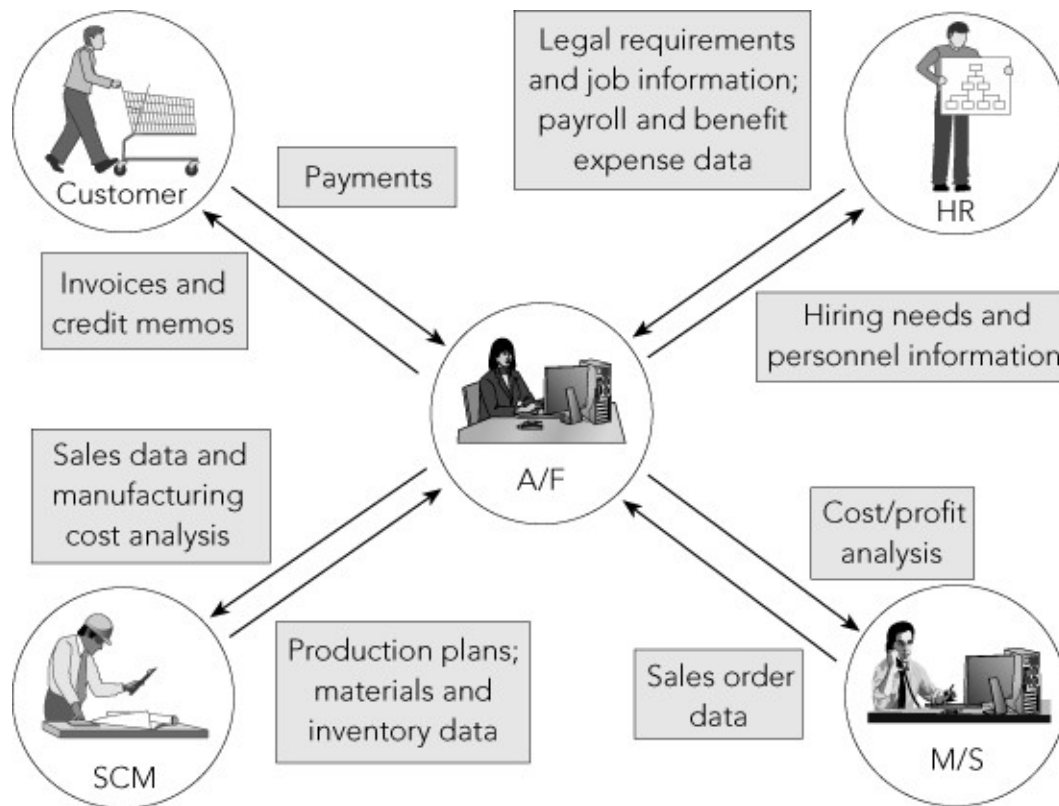
- Need information from all other functional areas
- **Inputs for M/S include:**
 - Customer data
 - Order data
 - Sales trend data
 - Per-unit cost
 - Travel expense company policy
- **Outputs for M/S include:**
 - Sales strategies
 - Product pricing
 - Employment needs

Supply Chain Management (SCM)



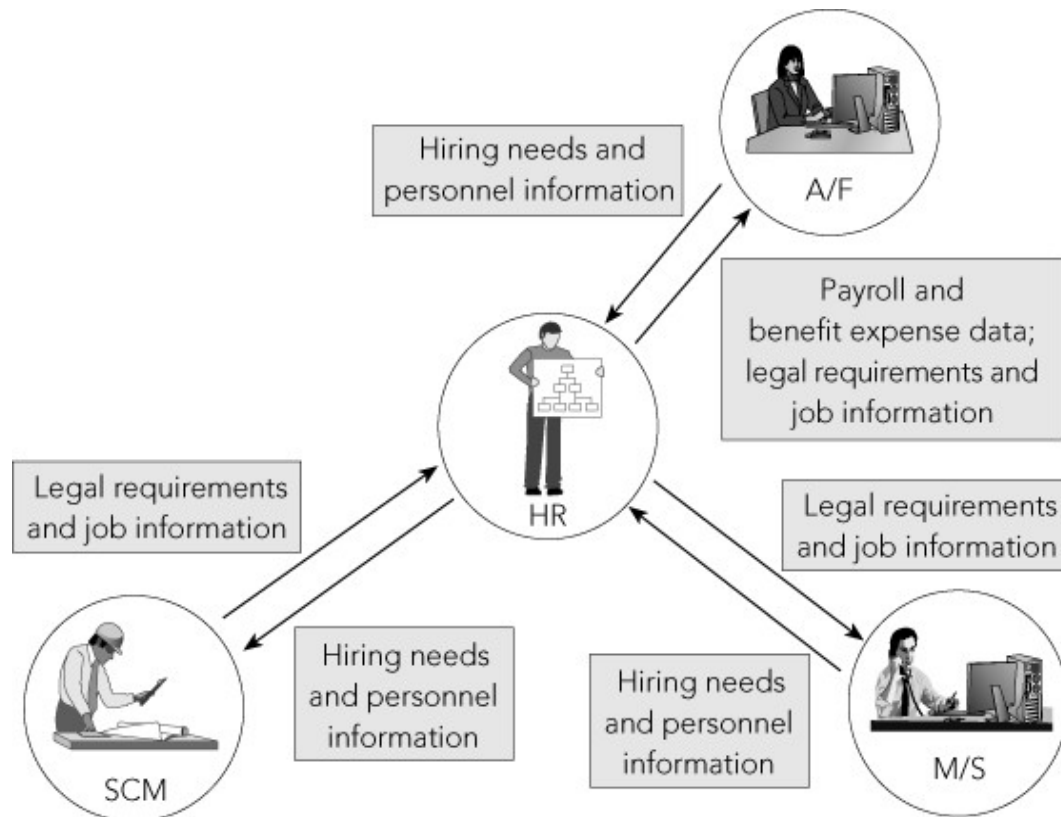
- Need information from all other functional areas
- **Inputs for SCM include:**
 - Product sales data
 - Production plans
 - Inventory levels
 - Layoff and recall company policy
- **Outputs for SCM include:**
 - Raw material orders
 - Packaging orders
 - Resource expenditure data
 - Production and inventory reports
 - Hiring information

Accounting and Finance (A/F)



- Need information from all other functional areas
- **Inputs for A/F include:**
 - Payments from customers
 - Accounts receivable data
 - Accounts payable data
 - Sales data
 - Production and inventory data
 - Payroll and expense data
- **Outputs for A/F include:**
 - Payments to suppliers
 - Financial reports
 - Customer credit data

Human Resources(HR)



- Need information from all other functional areas
- **Inputs for HR include:**
 - Personnel forecasts
 - Skills data
- **Outputs for HR include:**
 - Regulation compliance
 - Employee training and certification
 - Skills database
 - Employee evaluation and compensation

Popular ERP

- All ERPs are integrated information systems (IIS), but not all IIS are ERP systems.
 - SAP ERP
 - Oracle ERP Cloud
 - Microsoft Dynamics 365
 - NetSuite ERP
 - Odoo
 - Infor ERP



Summary

Definitions:

Enterprise Resource Planning (ERP) programs are software used by companies to manage information in every area of the business.

A **business function** is a **specialized area or department** within an organization that performs a specific set of related tasks or responsibilities to support the overall operation of the business.

A **business process** is a **sequence of interrelated tasks or activities** that produce a specific service or product to achieve a business goal often **crossing multiple functions**.

Information systems that share data between functional areas are called **Integrated Information Systems**

Summary

Basic functional areas:

1. Marketing and Sales,

- ☐ Sets product prices, promotes products through advertising and marketing, takes customer orders, supports customers, and creates sales forecasts

2. Supply Chain Management,

- ☐ Develops production plans, orders raw materials from suppliers, receives raw material, manufactures products, maintains facilities, and ships products to customers

3. Accounting and Finance,

- ☐ Financial accounting to provide summaries of operational data in managerial reports, controlling accounts, planning and budgeting, and cash-flow management

4. Human Resources

- ☐ Recruits, hires, trains, and compensates employees, ensures compliance with government regulations, and oversees the evaluation of employees

Summary

- **Employees working in one functional area need data from employees in other functional areas**
- **Functional area information systems should be integrated, so shared data are accurate and timely**
- **Managers think in terms of business processes that integrate the functional areas**
- **Need to share information between functions and functional areas**
- **ERP software provides this capability by means of a single common database**